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15 July 1982
SUPERSEDING
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4 December 1974

MILITARY SPECIFICATION
PROPELLANT M10
FOR 81MM MORTAR M205 PROPELLING CHARGE

This specification is approved for use by the U. S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope This specification covers M10 Propellant for use in M205 Mortar Increments for the following 81MM cartridges: HE, M374A3 and Smoke, WP, M375A3.

1.2 Classification. The propellant shall be M10 flake.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1376

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STANDARDS

MILITARY

MIL-STD-414 - Sampling Procedures and tables for Inspection by variables for percent defective

MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

2.1.2 Other government documents, drawings, and publications. The following documents and drawings form a part of this specification to the extent specified herein.

DRAWINGS (see 6.9)

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9280588 - Charge, Propelling, M205
9278773 - Container Assembly
9241291 - Cartridge, 81MM, HE M374A3
9294735 - Cartridge, 81MM, Smoke, WP, M375A3

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M10, conforming to the requirements of MIL-STD-652 and Drawing 9280588 and with a nominal thickness of 0.014 inches except that the maximum total volatiles content shall be 1.80 percent and the graphite glaze shall not exceed 0.2 percent.

3.2 Ballistic assessment requirements. The propellant, when fired in the M29 Mortar series and with the projectile (M374A1) (see 6.5) for which it was manufactured, shall comply with the following requirements at $70^{O}F$ and standard weapon conditions (see 6.11). The propellant shall be capable of assessment to a velocity of 885 feet per second (fps) with an expected average pressure not to exceed 9200 pounds per square inch (PSI) and a charge weight not to exceed 1750 grains. (see 6.14)

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3.3 Ballistic uniformity requirements. The propellant, when fired in the M29 Mortar series at the recommended charge weight and with the projectile (M374A1) (see 6.5) for which it was manufactured, shall comply with the following requirements (see 4.5.1.2).

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 5.0 fps.

3.3.2 Average Pressure At standard weapon conditions, the propellant shall not produce an average pressure greater than 9200 psi at 70°F.

3.3.3 Individual maximum pressure. The propellant shall not produce an individual pressure greater than 10,600 psi at 145°F when fired in a standard weapon (see 6.10).

3.4. Preproduction Inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in the contract.

3.5. Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of Inspections. Preproduction Sample Inspection and Quality Conformance inspection shall be conducted on this item.

4.3 Preproduction sample

4.3.1 Submission. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2. The preproduction samples shall consist of three (3) pilot lots, approximately 100 pounds each. A new sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity, loadability, or ballistic performance as determined by the Technical Agency (see 6.12).

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. A pilot lot which meets the requirements of the applicable drawings and specifications may be designated by the Contracting officer as the accepted preproduction sample. Until the preproduction sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the Contracting Officer.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

PRECAUTION - This specification covers sampling and testing of chemical toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2 Examination and testing

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652.

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4.4.2.2. Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the contracting officer.

4.4.2.4 Velocity uniformity (70°F) (see 3.3.1). If the product of the factor (0.73) times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3, and 4.5.1.4. If the retest and referral criteria do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.2). If any average pressure obtained during initial or subsequent production at 70°F, when corrected to standard conditions and recommended charge weight, exceeds the applicable requirement, apply retest and referral criteria in accordance with 4.5.1.3. and 4.5.1.4 If the retest and referral criteria do not apply, the lot shall be rejected.

4.4.2.6 Individual maximum pressure (initial production). If any individual pressure observed during the 145°F uniformity series, when corrected to standard conditions and recommended charge weight, exceeds 10,600 psi, the lot shall be rejected.

4.4.2.7 Individual maximum pressure (subsequent production) If any individual pressure observed during the 70°F uniformity series, when corrected to standard conditions and recommended charge weight, exceeds the 70°F individual maximum pressure (10,600 psi minus the quantity of 75 times the pressure - temperature coefficient established from initial production testing), the lot shall be tested at +145°F in accordance with 4.5.1.2.1. If any individual pressure at 145°F, when corrected to standard conditions and recommended charge weight, exceeds 10,600 psi, testing shall cease and the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments. This test shall be performed at a Government Proving Ground in accordance with the applicable acceptance test procedure using M374A3, HE cartridges inert loaded within 0.01 lb of an as fired weight of 9.05 lb and fired from M29 Mortar tube on a hard mount at an elevation of 800 mils. All charges shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges

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are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: -50°F, +5°F; 70°F, +2.5°F; 145°F, -5°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each round. Velocity coils will be used to measure the muzzle velocity of each round. The container assembly lot number of the test rounds shall also be recorded.

4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three or more groups of rounds, three rounds for each group loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. Three calibration rounds shall also be conditioned and fired at 70°F, prior to the test rounds. From the data obtained, a composite charge weight versus velocity versus pressure curve shall be established. A charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The result will be the recommended charge weight.

4.5.1.1.2 Constant Slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity versus pressure curve can be established from test results generated. This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as above for the charge slope method. The resultant will be the recommended charge weight.

4.5.1.2 Uniformity

4.5.1.2.1 Initial production. Thirty rounds shall be assembled with the calculated charge weight of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight

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(see 3.2), and to recommended charge weight conditions (i. e., the difference between as fired uniformity charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficients for ten lots shall be utilized to calculate a 70°F individual maximum pressure for use in subsequent production.

4.5.1.2.2 Subsequent production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds. The means and standard deviations of the velocity and pressure shall be calculated on results corrected to standard conditions and recommended charge weight. If any individual pressure exceeds the 70°F individual maximum pressure (see 4.4.2.7), ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (see 3.2) and recommended charge weight. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan (see 4.5.1.2.1).

4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds 40 grains of the charge weight used in uniformity firing, retest for charge establishment and uniformity at 70°F.

b. If for any reason, the proving ground considers that test conditions have detrimentally affected the test results, additional rounds as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exists or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 17.5 fps of the expected muzzle velocity.

c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

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e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X^2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed Level A.

6. NOTE

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions.

6.1.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 shall apply.

6.1.2. Contract data requirements One copy each of the Propellant Description sheet and the Acceptance Test Summary shall be forwarded to each office designated by the contracting officer. Copies of data item descriptions required by the contractor in connection with specific acquisition functions should be obtained from the procuring activity.

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6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3. Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Throughout production, if 3 consecutive lots fail on first test or a lapse in production of 90 days or more occurs, testing shall revert back to initial production plan (see 4.5.1.2.1).

6.4 Surveillance sample. Samples shall be forwarded to U. S. Army R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Test rounds. Test rounds shall be assembled in accordance with drawings 9241291, 9280588, 9278773 and 9294735.

6.6 Proving ground test summary

Temp	Sample Size		Test Results			
			Velocity (fps)		Pressure (psi)	
°F	Cal	Test	Avg	Std Dev	Ind Max	Max Avg
			Charge Establishment			
70	3	9	885	-	-	9,200
			Uniformity (Initial Production)			
-50	-	10	-	-	-	-
70	10	10	885	5.0	-	9,200
145	-	10	-	-	10,600	-
			Uniformity (Subsequent Production)			
70	10	10	885	5.0	see 4.5.1.2.2	9,200
145*	-	10	-	-	10,600	-

* To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.5.1.2.2 and 4.4.2.7)

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6.7 Retest. All original and retest data shall be submitted to U. S. Army R&D Command, ATTN: DRDAR-LCA-G, DRDAR-QAR-R Dover, New Jersey 07801.

6.8 Test anomalies. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc., shall be reported in detail on the applicable firing record.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.10 Individual Maximum Pressure Basis. The individual maximum pressure value for 145°F cited is based upon the weapon design pressure.

6.11 Standard weapon. Standard weapon and standard weapon conditions refer to the weapon model and conditions of the test at the time the master calibration series was fired.

6.12 Preproduction sample. The preproduction sample is intended to provide the contractor with the necessary latitude to establish the propellant granulation needed to meet the required ballistic parameters. The Technical Agency and the contractor should jointly select the one pilot lot which best represents the desired parameters. All future production lots should be of the same basic granulation as the accepted pilot lot.

6.13 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to U. S. ARMY Armament R&D Command, Dover, New Jersey 07801, ATTN: DRDAR-LCA-G for use in evaluating the propellant stockpile.

6.14 Velocity requirements. The velocity requirement in paragraph 3.2 reflects firing from a hard mount (armor plate) which is approximately 10 fps higher than velocities obtained from a soft mount (ground).

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